ZikShare

ACADEMIC VAULT

UNIZIK FRESHER EDITION | 2026

PHA 101: Introduction to Pharmacy - Drugs, Dosage Forms and the Pharmacist's Role

Orientation to pharmacy practice, drug nomenclature and pharmaceutical calculations

FACULTY PRIMER

Faculty Primer

Faculty of Pharmaceutical Sciences
Pharmaceutics & Pharmaceutical Technology | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

Free to download. No fee, no payment step, no watermark.

Independent study support | Not an official UNIZIK publication

START HERE

Why this pack exists

This pack orients new pharmacy students to the profession, explains drug nomenclature and dosage forms, and drills the pharmaceutical calculations that catch students out in early assessments, including percentage concentrations, dilutions and dose calculations.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Explain the pharmacist's roles in dispensing, counselling, compounding and pharmacovigilance.
- Distinguish chemical, generic and brand names and discuss bioequivalence.
- Classify dosage forms and explain how the route of administration affects onset and bioavailability.
- Perform percentage, dilution and dose calculations accurately with correct units.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

What pharmacy is for

Pharmacy is the science and practice of preparing and using medicines safely and effectively.

- Dispensing: verifying the prescription, preparing and labelling the medicine.
- Counselling: explaining dose, frequency, duration, storage and side effects to the patient.
- Compounding: extemporaneous preparation where no commercial product fits.
- Pharmacovigilance: detecting, assessing and reporting adverse drug reactions.
- Regulation: controlled drugs, prescription requirements and record keeping.
- Medication reconciliation and adherence support reduce avoidable harm.

> **The pharmacist's check**
> Right patient, right drug, right dose, right route, right time and right documentation. Any doubt is a reason to stop and verify, not to guess.

Names, dosage forms and bioavailability

Names and formulations are not trivia; they determine how a medicine behaves in the body.

- Chemical name describes structure; generic name is the internationally accepted non-proprietary name; brand name is the manufacturer's trade name.
- Bioavailability is the fraction of an administered dose that reaches systemic circulation unchanged.
- Bioequivalence means two products deliver the same rate and extent of absorption within accepted limits.
- Oral solid forms: tablets, capsules, granules. Modified-release designs alter release rate.
- Liquid forms: solutions, suspensions, emulsions; shake suspensions before measuring.
- Parenteral, topical, rectal, inhaled and ophthalmic routes each have distinct advantages and risks.

Calculations without panic

Set every calculation out in one consistent format: known, required, conversion, equation, answer with units.

- Percentage weight in volume and weight in weight are not interchangeable; read what the label states.
- Dilution: C1 x V1 = C2 x V2, with concentrations in the same unit.
- Dose by body weight: dose in mg per kg multiplied by weight in kg, then checked against the maximum adult dose.
- Dose by body surface area requires the stated formula and correctly converted height and weight.
- Alligation solves mixing problems between two strengths.
- Always sanity-check: an answer that implies 40 tablets per dose is almost certainly wrong.

> **Units decide safety**
> Milligrams and micrograms differ by a factor of 1000. A decimal slip can be fatal. Convert to the same unit before calculating and write the unit at every step.

From molecule to medicine

A drug is a chemical producing a biological effect; a medicine is a formulated product that delivers it safely. The difference is pharmaceutical science: solubility, stability, particle size, excipients, packaging and storage determine whether the active ingredient reaches the patient in usable form.

Pharmacokinetics describes what the body does to the drug - absorption, distribution, metabolism, excretion. Pharmacodynamics describes what the drug does to the body. Most clinical questions about dosing are kinetics questions in disguise.

Bioavailability is the proportion of a dose reaching systemic circulation unchanged. Intravenous administration is 100 per cent by definition; oral bioavailability is reduced by incomplete absorption and first-pass metabolism, which is why oral and intravenous doses differ.

Formulation and dosage forms

Tablets contain active ingredient plus excipients: diluents, binders, disintegrants, lubricants and coatings. Each has a function and each can cause problems - a binder that slows disintegration delays onset; a coating may protect the stomach or protect the drug from it.

Modified-release systems control the rate of release. They improve convenience and steady plasma levels but introduce risk: crushing a sustained-release tablet releases the whole dose at once, which can be dangerous. Counselling on this is a pharmacist's responsibility.

Sterile products - injections, eye drops, infusions - require aseptic manufacture and endotoxin control. Their preparation is regulated because contamination cannot be seen.

> **The dose makes the poison**
> Paracetamol is safe at therapeutic dose and causes liver failure in overdose. Every counselling conversation about a common medicine should include the maximum daily dose.

The pharmacist's professional role

Pharmacists dispense, but their value is in validation, counselling and monitoring. Checking interactions, contraindications, duplicate therapy and appropriate dosing prevents harm that prescribing alone does not catch.

Counselling must be specific: what the medicine is for, how and when to take it, what to do about a missed dose, likely side effects, and when to seek help. Teach-back confirms understanding, and written instructions in a language the patient reads are more reliable than a hurried explanation.

Ethics and law govern controlled substances, record keeping, substitution and advertising. Selling antibiotics without prescription, dispensing beyond competence, or failing to report adverse reactions are professional failures with public health consequences - antimicrobial resistance being the clearest.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. Calculating a paediatric dose

A child weighs 18 kg and is prescribed 15 mg/kg/day in three doses. The suspension is 125 mg/5 mL. Find the volume per dose.

1. Daily dose: 15 x 18 = 270 mg.
2. Per administration: 270 / 3 = 90 mg.
3. Concentration: 125 mg in 5 mL, so 25 mg per mL.
4. Volume: 90 / 25 = 3.6 mL.
5. Check against the maximum for the child's age and weight.
6. Show the working and have it independently checked.

> **What to carry forward**
> Convert to a common unit, calculate, check against a maximum, show the working.

2. Counselling on a new antibiotic

A patient collects a course for a chest infection.

1. Confirm identity, prescription and allergies.
2. State the purpose in plain language.
3. Give dose, frequency, duration and the reason to complete the course.
4. Mention common side effects and what to do about them.
5. State when to seek help: rash, breathlessness, worsening fever.
6. Ask the patient to repeat the key points and provide written instructions.

> **What to carry forward**
> Purpose, dose, duration, side effects, red flags, teach-back.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Explain the four processes of pharmacokinetics with one factor affecting each.

20 marks

1. Absorption: formulation, food, gastric emptying.
2. Distribution: protein binding, lipid solubility, perfusion.
3. Metabolism: liver function, enzyme induction and inhibition.
4. Excretion: renal function and urine pH.
5. Conclude that these determine dose and interval.

Q2. Why must modified-release tablets not be crushed?

10 marks

1. Explain the release mechanism.
2. Describe dose dumping and its risk.
3. Note exceptions where crushing is licensed.
4. Relate to counselling practice.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Pharmacokinetics. What the body does to a drug: absorption, distribution, metabolism, excretion.

Pharmacodynamics. What a drug does to the body.

Bioavailability. The proportion of a dose reaching systemic circulation unchanged.

First-pass metabolism. Liver breakdown of an oral drug before it reaches circulation.

Excipient. An inactive ingredient included for a specific function.

Disintegrant. An excipient helping a tablet break apart in the gut.

Modified-release. A formulation controlling the rate or site of release.

Dose dumping. Rapid release of the full dose from a damaged modified-release product.

Contraindication. A circumstance in which a medicine should not be used.

Adverse drug reaction. A harmful response at normal doses.

Aseptic preparation. Compounding sterile products under conditions preventing contamination.

Antimicrobial resistance. Microorganisms surviving drugs that previously killed them.

Your four-week plan

Week 1 - Kinetics

ADME.

- Define the four processes.
- Explain bioavailability.
- Compare oral and intravenous dosing.

Week 2 - Formulation

Dosage forms.

- List excipients and functions.
- Explain modified release.
- Describe why crushing is dangerous.

Week 3 - Calculations

Accuracy.

- Solve ten dose calculations.
- Check each against a maximum.
- Practise paediatric weight-based dosing.

Week 4 - Practice

Counselling.

- Role-play three counselling scenarios.
- Write plain-language instructions.
- List when to refer.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any introduction to pharmaceutics covering formulation and dosage forms.
- A clinical pharmacy text on counselling and dose calculation.
- Your national drug regulation guidance.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. A prescription requires 250 mg and the stock is a 5 percent w/v solution. What volume is needed?

Answer: Five percent w/v is 5 g per 100 mL, or 50 mg per mL. 250 mg therefore requires 5 mL.

2. Dilute 100 mL of a 20 percent solution to 500 mL. What is the final concentration?

Answer: C1V1 = C2V2 gives 20 x 100 = C2 x 500, so C2 = 4 percent.

3. What is bioequivalence?

Answer: The absence of a significant difference in the rate and extent to which the active ingredient becomes available at the site of action between two products.

4. Why must a suspension be shaken before use?

Answer: The dispersed particles settle, so shaking redistributes them and ensures the measured dose is representative.

5. A child weighs 20 kg and the dose is 10 mg/kg per day in two divided doses. What is each dose?

Answer: The daily dose is 200 mg, divided into two doses of 100 mg each.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] List six roles of a pharmacist and one example of each.
- [ ] Explain chemical, generic and brand names in one sentence each.
- [ ] Make a table of dosage forms with two advantages and two disadvantages.
- [ ] Solve ten percentage and dilution calculations showing all units.
- [ ] Memorise the right-patient checking list.
- [ ] Practise weighing and measuring in the laboratory with a supervisor.
- [ ] Read the current NAFDAC or PCN guidance on dispensing and write five key points.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

> **More from the ZikShare Academic Vault**
> Find faculty primers, study-skills guides, money and career packs, and digital-skills workbooks in the ZikShare Library. New packs are added every semester.

ZikShare Academic Vault | Faculty Primer | 100L | 2026 Fresher Edition

YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

PHA 101: Introduction to Pharmacy - Drugs, Dosage Forms and the Pharmacist's Role